Submission for BAW-INEQBASIC

OTIS (internal use)

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Example (Australia Training, added by Leonardo Wang, 0♣)**.** Suppose that x_1, x_2, $\ldots$, x_n are positive real numbers which satisfy $x_1x_2\ldots x_n = 1$. If $r > s > 0$, prove that

$$x_1^r + x_2^r + \cdots + x_n^r \geq x_1^s + x_2^s + \cdots + x_n^s.$$

21AUSTRAINING

Walkthrough. There are two approaches here. The first approach is by Muirhead:

(a) Homogenize the inequality using $x_1x_2\ldots x_n = 1$.

(b) Solve using Muirhead.

Another approach is to directly compare with power mean:

(c) Use power mean to compare the r'th and s'th power mean.

(d) Prove unconditionally that

$$\frac{x_1^r + \cdots + x_n^r}{n} \geq \frac{x_1^s + \cdots + x_n^s}{n} \cdot (x_1 \ldots x_n)^k$$

for some real number $k > 0$, and conclude.

Example (Taiwan TST Quiz 2014, 0♣)**.** For $a, b, c > 0$ prove that

$$3(a+b+c) \geq 8\sqrt[3]{abc} + \sqrt[3]{\frac{a^3+b^3+c^3}{3}}.$$

14TWNQ1J1

Walkthrough. This inequality isn't particularly sharp, so the obvious thing works:

(a) Eliminate the cube roots using power mean, so that it suffices to prove

$$(a+b+c)^3 \geq a^3 + b^3 + c^3 + 24abc.$$

(b) Check that this is true.

Example (0♣)**.** If $abcd = 1$ for $a, b, c, d > 0$, prove that

$$a^4b + b^4c + c^4d + d^4a \geq a + b + c + d.$$

Z1422B10

Walkthrough. There are two possible solutions I know of, one by Hölder and one by AM-GM. I find the latter much more natural. (And no, Muirhead doesn't apply, because it's cyclic not symmetric.)

(a) Homogenize the inequality to eliminate the condition (while keeping the inequality fifth-degree).

(b) Fill in the blanks in the following AM-GM:

$$? \cdot a^4b + ? \cdot b^4c + ? \cdot c^4d + ? \cdot d^4a \geq a^2bcd.$$

(c) Cyclically sum to finish.

Depending on how you did this, the number 51 might appear.

Example (IMO 2001/2, 0♣). Let a, b, c be positive reals. Prove that

$$\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ca}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1.$$

01IMO2

Walkthrough. There are a few ways to set up, but the general idea is to use Hölder in the form

$$\left(\sum_{\text{cyc}} \frac{a}{\sqrt{a^2+8bc}}\right)^p \left(\sum_{\text{cyc}} ?\right)^q \geq \left(\sum_{\text{cyc}} ?\right)^{p+q}$$

for some choice of weights p and q to eliminate the radicals and get a polynomial inequality.

(a) Pick a choice of weights $p, q > 0$ eliminate the radicals.

(b) Decide on values to fill in the ? above. (You probably want to eliminate the denominator, i.e. the left sum should be some multiple of $a^2 + 8bc$.)

(c) Try to prove the resulting inequality. Depending on what choices you made in (a) or (b), this may be relatively easy, or it may be impossible (because the inequality may not even be true.)

Example (IMO 2005/3, 0♣). Let $x, y, z > 0$ satisfy $xyz \geq 1$. Prove that

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0.$$

05IMO3

Walkthrough.

(a) By subtracting 3 from both sides and manipulating, rewrite the inequality in a form that does not have any minus signs, and for which the fractions have constant numerator.

(b) Show that we can assume WLOG that $xyz = 1$.

(c) Homogenize the resulting inequality.

(d) EXPAND! (Or try to find a nicer solution if you're so inclined.)

Practice problems

Instructions: Solve [42♣]. If you have time, solve [50♣].

It's psychosomatic. You need a lobotomy. I'll get a saw.

Dr. Calvin in *Calvin and Hobbes*

Problem 1 (IMO 1983/6, added by Aditya Ghumre, 3♣)

Prove that if a, b, c are the side lengths of a triangle, then

$$a^2b(a-b)+b^2c(b-c)+c^2a(c-a)\geq 0.$$

83IMO6

Proof. We let $(a,b,c)=(y+z,z+x,x+y)$ for x, y, $z>0$, so the problem boils down to proving the following:

$$\sum_{\text{cyc}}(x+y)^2(y+z)(x-z)\geq 0$$

After some brutal expansion, we obtain:

$$xy^3+yz^3+zx^3\geq x^2yz+xy^2z+xyz^2$$

Dividing throughout by xyz we obtain

$$\frac{y^2}{z}+\frac{z^2}{x}+\frac{x^2}{y}\geq x+y+z$$

Which is true by Engel form Cauchy. □

Problem 2 (USAMO 1983/2, 2♣)

Prove that the roots of

$$x^5+ax^4+bx^3+cx^2+dx+e=0$$

cannot all be real if $2a^2<5b$.

83AMO2

Proof. Let $P(x)=x^5+ax^4+bx^3+cx^2+d$. Compute $P'''(x)=60x^2+24ax+6b$. By Rolle's theorem, if P has all 5 real roots, then P''' has two real roots, which means its discriminant is nonnegative. This gives us $2a^2\geq 5b$, which means that $2a^2<5b$ if all the roots are not all real. □

Problem 3 (Kyle Wu, 2♣)

For positive real numbers a, b, and c with $a+b+c=1$, prove that

$$\frac{b^2+c^2}{1+a}+\frac{c^2+a^2}{1+b}+\frac{a^2+b^2}{1+c}\geq\frac{1}{2}.$$

KYLEWU

Proof. The problem is equivalent to showing

$$\frac{b^2}{2a+b+c}+\frac{c^2}{2a+b+c}+\frac{c^2}{2b+a+c}+\frac{a^2}{2b+a+c}+\frac{a^2}{2c+a+b}+\frac{b^2}{2c+a+b}\geq\frac{1}{2}$$

Which follows immediately by Titu's Lemma. □

Problem 4 (USAMO 1978/1, 3♣)

Given that a, b, c, d, e are real numbers such that $a+b+c+d+e=8$ and $a^2+b^2+c^2+d^2+e^2=16$, determine the maximum value of e.

78AMO1

Proof. By Cauchy-Schwarz, we have:

$$(a^2+b^2+c^2+d^2)(4)\geq(a+b+c+d)^2 \implies 4(16-e^2)\geq(8-e)^2$$

giving $e(5e-16)\leq 0$ so $\max\{e\}=\frac{16}{5}$. □

Problem 5 (3♣)

For $a, b, c>0$ prove that

$$\frac{c^2+ab}{a+b}+\frac{a^2+bc}{b+c}+\frac{b^2+ca}{c+a}\geq a+b+c.$$

ZB2F3EE8

Problem 6 (Shortlist 1998 A3, added by Leonardo Wang, 3♣)

Let x, y, z be positive real numbers which satisfy $xyz=1$. Show that

$$\sum_{\text{cyc}}\frac{x^3}{(1+y)(1+z)}\geq\frac{3}{4}.$$

98SLA3

Problem 7 (JMO 2012/3, 3♣)

For $a, b, c>0$ prove that

$$\frac{a^3+3b^3}{5a+b}+\frac{b^3+3c^3}{5b+c}+\frac{c^3+3a^3}{5c+a}\geq\frac{2}{3}(a^2+b^2+c^2).$$

12JMO3

Problem 8 (USAMO 1997/5, 3♣)

If $a, b, c>0$ prove that

$$\frac{1}{a^3+b^3+abc}+\frac{1}{b^3+c^3+abc}+\frac{1}{c^3+a^3+abc}\leq\frac{1}{abc}.$$

97AMO5

Problem 9 (IMO 2000/2, 3♣)

Let a, b, c be positive real numbers with $abc = 1$. Show that

$$\left(a - 1 + \frac{1}{b}\right)\left(b - 1 + \frac{1}{c}\right)\left(c - 1 + \frac{1}{a}\right) \leq 1.$$

00IMO2

Proof. Let $a = \frac{x}{y}$, $b = \frac{y}{z}$, $c = \frac{z}{x}$ so that the problem reduces to proving:

$$\sum_{\text{cyc}} \left(\frac{x}{y} - 1 + \frac{z}{y}\right) \leq 1$$

Expanding this gives us:

$$x^3 + y^3 + z^3 + 3xyz \geq x^2y + y^2z + z^2x$$

Which is true by Schur's. □

Problem 10 (Lin Liu, 3♣)

Prove that for any $a, b, c > 0$,

$$\left(\frac{a+b+c}{3}\right)\left(\sum_{\text{cyc}} \frac{b^{3/2}}{\sqrt{a}}\right) \geq \sum_{\text{cyc}} a(2b - a).$$

LINLIU

Problem 11 (Latvia Baltic Way TST 2019, added by Kims Georgs Pavlovs, 9♣)

Prove that for all positive real numbers a, b, c with $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$ the following inequality holds:

$$3(ab + bc + ca) + \frac{9}{a+b+c} \leq \frac{9abc}{a+b+c} + 2(a^2 + b^2 + c^2) + 1.$$

Z148E702

Problem 12 (Mathematical Reflections O-459, 2♣)

Let a, b, x be real numbers such that

$$(4a^2b^2 + 1)x^2 + 9(a^2 + b^2) \leq 2018.$$

Prove that

$$20(4ab + 1)x + 9(a + b) \leq 2018.$$

MRO459

Problem 13 (Iran 1998, 5♣)

Let x, y, z be real numbers greater than 1. Prove that if $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 2$ then

$$\sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1} \leq \sqrt{x+y+z}.$$

98IRN

Problem 14 (CGMO 2012/1, 3♣)

Let n be a positive integer, and let a_1, a_2, $\ldots$, a_n be nonnegative real numbers. Show that

$$\frac{1}{1+a_1} + \frac{a_1}{(1+a_1)(1+a_2)} + \cdots + \frac{a_1 a_2 \cdots a_{n-1}}{(1+a_1)(1+a_2)\cdots(1+a_n)} \leq 1.$$

Moreover, determine all the cases of equality.

12CGMO1

Required Problem 15 (IMO 2012/2, 5♣)

Let a_2, a_3, $\ldots$, a_n be positive reals with product 1, where $n \geq 3$. Show that

$$(1+a_2)^2(1+a_3)^3 \ldots (1+a_n)^n > n^n.$$

12IMO2

Proof. By AM-GM, we have:

$$(1+a_k)^k = \left(\frac{1}{k-1} + \cdots + \frac{1}{k-1} + a_k\right)^k \geq \frac{k^k}{(k-1)^{k-1}} a_k$$

Giving:

$$\begin{aligned}\prod (1+a_k)^k &\geq \prod \left(\frac{k^k}{(k-1)^{k-1}} a_k\right) \\ &= \prod \left(\frac{k^k}{(k-1)^{k-1}}\right) \\ &= n^n\end{aligned}$$

We prove that equality cannot be achieved. From the AM-GM, equality is achieved if and only if $a_k = \dfrac{1}{k-1}$ for $k = 2, 3, \ldots, n$. However this is a contradiction as $a_2 a_3 \ldots a_n = \frac{1}{(n-1)!} \neq 1$. So the inequality is strict, and we're done! □

Required Problem 16 (ELMO 2003, Po-Ru Loh, 5♣)

Let $x, y, z > 1$ be real numbers such that

$$\frac{1}{x^2-1} + \frac{1}{y^2-1} + \frac{1}{z^2-1} = 1.$$

Prove that

$$\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} \leq 1.$$

03ELMO4

Proof. We set:

$$a = \frac{1}{x^2-1} \implies x = \sqrt{\frac{a+1}{a}}$$

Define b and c analogously. If we define a function:

$$f(x) = \frac{1}{\sqrt{\frac{x+1}{x}} + 1} = (x^2+x)^{\frac{1}{2}} - x,$$

then it suffices to show that $f(a)+f(b)+f(c) \le 1 = 3f\left(\frac{a+b+c}{3}\right)$ which is the same as Jensen's. This means that showing f is concave on $(0,\infty)$ solves the problem. We compute:

$$f''(x) = -\frac{1}{4(x^2+x)^{\frac{3}{2}}} < 0 \text{ for } x \in (0,\infty)$$

And we're done! □

Problem 17 (Shortlist 2004 A5, 9♣)

If a, b, c are three positive real numbers such that $ab+bc+ca=1$, prove that

$$\sqrt[3]{\frac{1}{a}+6b} + \sqrt[3]{\frac{1}{b}+6c} + \sqrt[3]{\frac{1}{c}+6a} \le \frac{1}{abc}.$$

04SLA5

Required Problem 18 (IMO 2004/4, 5♣)

Let $n \ge 3$ be an integer and t_1, t_2, ..., t_n positive real numbers such that

$$n^2+1 > (t_1+t_2+\cdots+t_n)\left(\frac{1}{t_1}+\frac{1}{t_2}+\cdots+\frac{1}{t_n}\right).$$

Show that t_i, t_j, t_k are the sides of a triangle for all i, j, k with $1 \le i < j < k \le n$.

04IMO4

Proof. Assume, for the sake of contradiction, that there exists some t_a, t_b, and t_c with $t_a \ge t_b + t_c$. Then, we observe:

$$\begin{aligned}
\left(\sum_i t_i\right)\left(\sum_j \frac{1}{t_j}\right) &= n + \sum_{i<j}\left(\frac{t_i}{t_j}+\frac{t_j}{t_i}\right) \\
&= n + \left(\sum_{\substack{i<j \\ \{i,j\}\notin\{\{a,b\},\{a,c\}\}}} \left(\frac{t_i}{t_j}+\frac{t_j}{t_i}\right)\right) + \left(\frac{t_a}{t_b}+\frac{t_b}{t_a}\right) + \left(\frac{t_a}{t_c}+\frac{t_c}{t_a}\right) \\
&\ge n + 2\left(\binom{n}{2} - 2\right) + \left(\frac{t_a}{t_b}+\frac{t_b}{t_a}+\frac{t_a}{t_c}+\frac{t_c}{t_a}\right) \\
&= n + n^2 - n - 4 + \left(\frac{t_a}{t_b}+\frac{t_b}{t_a}+\frac{t_a}{t_c}+\frac{t_c}{t_a}\right) \\
&= n^2 - 4 + \left(\frac{t_a}{t_b}+\frac{t_b}{t_a}+\frac{t_a}{t_c}+\frac{t_c}{t_a}\right)
\end{aligned}$$

We note, however, that:

$$\begin{aligned}\frac{t_a}{t_b}+\frac{t_b}{t_a}+\frac{t_a}{t_c}+\frac{t_c}{t_a} &= \left(\frac{t_b+t_c}{t_a}+\frac{t_a(t_b+t_c)}{t_bt_c}\right)\\ &= (t_b+t_c)\left(\frac{1}{t_a}+\frac{t_a}{t_bt_c}\right)\\ &\overset{\text{AM-GM}}{\geq} (t_b+t_c)\left(\frac{1}{t_a}+\frac{4t_a}{(t_b+t_c)^2}\right)\\ &= \left(\frac{t_b+t_c}{t_a}\right)+\left(\frac{t_a}{t_b+t_c}\right)+\frac{3t_a}{t_b+t-c}\\ &\overset{\text{AM-GM}}{\geq} 2+\frac{3t_a}{t_b+t_c}\\ &\geq 5\end{aligned}$$

Which gives us our required contradiction, solving the problem. □

Problem 19 (Russia 2018, added by Prithuj Sarkar, 3♣)

Let $n \geq 2$ and let $x_1, x_2, \ldots, x_n$ be positive real numbers. Prove that

$$\frac{1+x_1^2}{1+x_1x_2}+\frac{1+x_2^2}{1+x_2x_3}+\cdots+\frac{1+x_n^2}{1+x_nx_1}\geq n.$$

18RUS112

Proof. Using AM-GM on the entire expression, we get that:

$$\frac{1+x_1^2}{1+x_1x_2}+\frac{1+x_2^2}{1+x_2x_3}+\cdots+\frac{1+x_n^2}{1+x_nx_1}\geq n\sqrt[n]{\frac{(1+x_1^2)(1+x_2^2)\ldots(1+x_n^2)}{(1+x_1x_2)(1+x_2x_3)\ldots(1+x_nx_1)}}$$

Therefore, it suffices to show that:

$$\prod_{i=1}^{n}(1+x_i^2)\geq\prod_{i=1}^{n}(1+x_ix_{i+1})$$

where indices are taken modulo n. This follows immediately by Cauchy-Schwarz on all $(1+x_i^2)(1+x_{i+1}^2)$. □

Problem 20 (Latvia Baltic Way TST 2020, added by Kims Georgs Pavlovs, 3♣)

Prove that for positive reals a, b, c satisfying $a+b+c=3$ the following inequality holds:

$$\frac{a}{1+2b^3}+\frac{b}{1+2c^3}+\frac{c}{1+2a^3}\geq 1.$$

Z93A61C5

Problem 21 (IMO 1995/2, 5♣)

For positive real numbers a, b, c satisfying $abc=1$, prove that

$$\frac{1}{a^3(b+c)}+\frac{1}{b^3(c+a)}+\frac{1}{c^3(a+b)}\geq\frac{3}{2}.$$

95IMO2

Proof. Let $a = \frac{1}{x}$, $b = \frac{1}{y}$, $c = \frac{1}{z}$ such that we again have $xyz = 1$, and the problem reduces to proving:

$$\frac{x^2}{y+z} + \frac{y^2}{x+z} + \frac{z^2}{x+y} \geq \frac{3}{2}$$

By Cauchy-Schwarz, we have:

$$((y+z)+(x+z)+(x+y))\left(\frac{x^2}{y+z} + \frac{y^2}{x+z} + \frac{z^2}{x+y}\right) \geq (x+y+z)^2$$

Which gives:

$$\frac{x^2}{y+z} + \frac{y^2}{x+z} + \frac{z^2}{x+y} \geq \frac{x+y+z}{2}$$

However it's obvious that $x+y+z \geq 3\sqrt[3]{xyz} = 3$ so we're done. □

Required Problem 22 (MOP 2011 R4.2, 9♣)

For positive real numbers a, b, c with $a+b+c=3$ prove that

$$\sqrt{\frac{a^3+b^3}{a+b}} + \sqrt{\frac{b^3+c^3}{b+c}} + \sqrt{\frac{c^3+a^3}{c+a}} + 9\sqrt[3]{abc} \leq 12.$$

11MOPR42

Proof. We note this is equivalent to proving:

$$\left(\sum_{\text{cyc}} \sqrt{a^2-ab+b^2}\right) + 9\sqrt[3]{abc} \leq 12.$$

By AM-GM, we have:

$$\sqrt[3]{abc} \leq \frac{\sqrt{ab}+\sqrt{bc}+\sqrt{ca}}{3}$$

Giving us:

$$\left(\sum_{\text{cyc}} \sqrt{a^2-ab+b^2}\right) + 9\sqrt[3]{abc} \overset{\text{AM-GM}}{\leq} \left(\sum_{\text{cyc}} \sqrt{a^2-ab+b^2}\right) + 3(\sqrt{ab}+\sqrt{bc}+\sqrt{ca})$$

Cauchy-Schwarz gives:

$$\begin{aligned}(\sqrt{a^2-ab+b^2} + 3\sqrt{ab}) &\leq \sqrt{(a^2-ab+b^2+3ab)(1+3)} \\ &= 2(a+b)\end{aligned}$$

Therefore, summing cyclically, we get the original expression is $\leq 4(a+b+c) = 12$, as desired. □

Problem 23 (Evan's driving test 2012, 2♣)

Prove that if $a, b, c > 0$ satisfy $a^3+b^3+c^3 = 3$ then

$$a^3(b+c) + b^3(c+a) + c^3(a+b) \leq 6.$$

Z2606BB8

Remark (Etymology)**.** I remember this problem all my life because I saw it on AoPS right before my driver's license test while surfing on my phone. The good news is that I solved it in my head during that driving test. The bad news is that it caused me to run a red light and not even notice until my examiner grilled me at the end of said driving test. (I failed my second try as well by running a stop sign and colliding with the curb. I hate driving so much.)

Problem 24 (Putnam 2003, added by Louis Qiu, 3♣)

Find the minimum value of

$$|\sin x + \cos x + \tan x + \cot x + \sec x + \csc x|$$

for real numbers x.

03PTNMA3

Problem 25 (USAMO 2002/2, 5♣)

Let ABC be a triangle such that

$$\left(\cot\frac{A}{2}\right)^2 + \left(2\cot\frac{B}{2}\right)^2 + \left(3\cot\frac{C}{2}\right)^2 = \left(\frac{6s}{7r}\right)^2,$$

where s and r denote its semiperimeter and its inradius, respectively. Prove that triangle ABC is similar to a triangle T whose side lengths are all positive integers with no common divisors and determine these integers.

02AMO2

Problem 26 (Japan 2016, added by Nico Young, 3♣)

Suppose that a, b, c, d are real numbers (not necessarily positive) such that

$$\begin{aligned}(a+b)(c+d) &= 2,\\ (a+c)(b+d) &= 3,\\ (a+d)(b+c) &= 4.\end{aligned}$$

Find the minimum possible value of $a^2+b^2+c^2+d^2$.

16JPNP7

Problem 27 (Hungary-Israel 2007/2, added by Paul Dao, 3♣)

Let a, b, c, d be real numbers such that

$$\begin{aligned}a^2 &\le 1\\ a^2+b^2 &\le 5\\ a^2+b^2+c^2 &\le 14\\ a^2+b^2+c^2+d^2 &\le 30.\end{aligned}$$

all hold. Prove that $a+b+c+d \le 10$.

ZB579ADD

Problem 28 (Balkan Shortlist 2024 A3, added by Paul Dao, 3♣)

Find all triples (a, b, c) of positive real numbers satisfying

$$11bc - 36b - 15c = abc$$
$$12ca - 10c - 28a = abc$$
$$13ab - 21a - 6b = abc.$$

ZBEEE7BC